Large-Scale Supermarket Sales and Customer Behaviour Analysis

Data Analytics Capstone Project

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Educational analysis using synthetic supermarket data

1. Business Problem

Why analyse supermarket sales?

Retail managers need clear information about revenue, branches, products, and customer spending.

Without organized analysis, comparing performance and planning inventory or promotions can be difficult.

This project demonstrates how Python and SQL can turn transaction records into useful summaries.

Goal: provide evidence-informed recommendations while recognizing the limits of synthetic data.

2. Dataset Overview

Four related CSV files stored in the project data/ folder

50,000
transactions

20
products

5
branches

5,000
customer records

The data is synthetic and was created for learning and demonstration.

It does not represent actual supermarket revenue or real customer behaviour.

3. Data Cleaning and Preparation

Reviewed data quality across completeness, validity, uniqueness, consistency, accuracy, and integrity.

Checked duplicate transaction IDs and missing essential values.

Converted transaction dates to a date format and filtered invalid quantities or prices.

Recalculated sales amount from quantity × unit price.

Loaded data into SQLite to support SQL-based analysis.

4. Tools and Methods

Python and pandas — loading, cleaning, grouping, and summarizing data.

NumPy — numerical operations.

SQLite and SQL — joins, grouping, HAVING, and window functions.

Matplotlib — charts for branch, product, and time-based comparisons.

SciPy — Welch independent two-sample t-test comparing Nairobi and Mombasa transaction values.

5. Overall Sales Scorecard

Results transcribed from the 02_eda.ipynb output

KES 38,292,565
Total revenue

50,000
Transactions

274,143
Units sold

KES 765.85
Average transaction value

6. Branch Performance

Revenue is relatively similar across the five synthetic branches

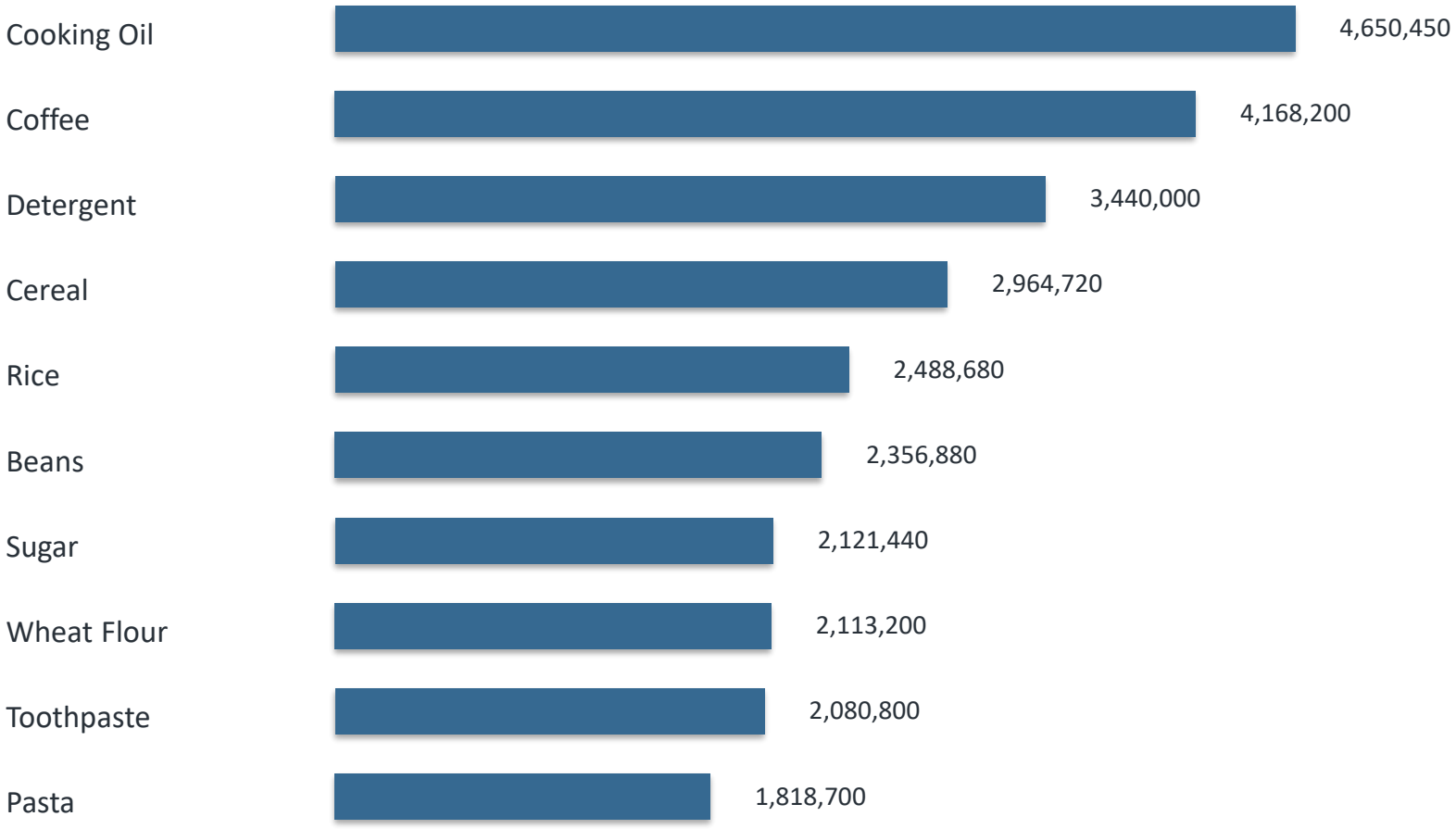


Highest: Kisumu (KES 7,756,715). Lowest: Eldoret (KES 7,541,125).

Difference: KES 215,590; interpret cautiously because the dataset is synthetic.

7. Top Products by Revenue

Top 10 products reported by the exploratory analysis



Key insight

Cooking Oil leads the listed products at KES 4,(

8. Recommendations and Limitations

Monitor monthly revenue and compare branch performance over time.

Use product-level results to guide further inventory and promotion analysis, not as a standalone stocking rule.

Validate patterns with real sales data before making budget or operational decisions.

Limitation: data is synthetic; observed branch and product differences are illustrative, not real-world evidence.

Statistical test: report the actual t-statistic, p-value, and interpretation from 03_analysis.ipynb before final submission.

Statistical test: A Welch independent two-sample t-test comparing Nairobi and Mombasa produced a t-statistic of 0.2944 and a p-value of 0.7685. Since the p-value exceeds 0.05, we failed to reject the null hypothesis. The test found insufficient evidence of a difference in average transaction values between the two branches in this synthetic dataset.

9. Conclusion

This project demonstrates an end-to-end analytics workflow: cleaning data, querying with S

The most important next step is to validate any business recommendations using real operational data.

Thank you